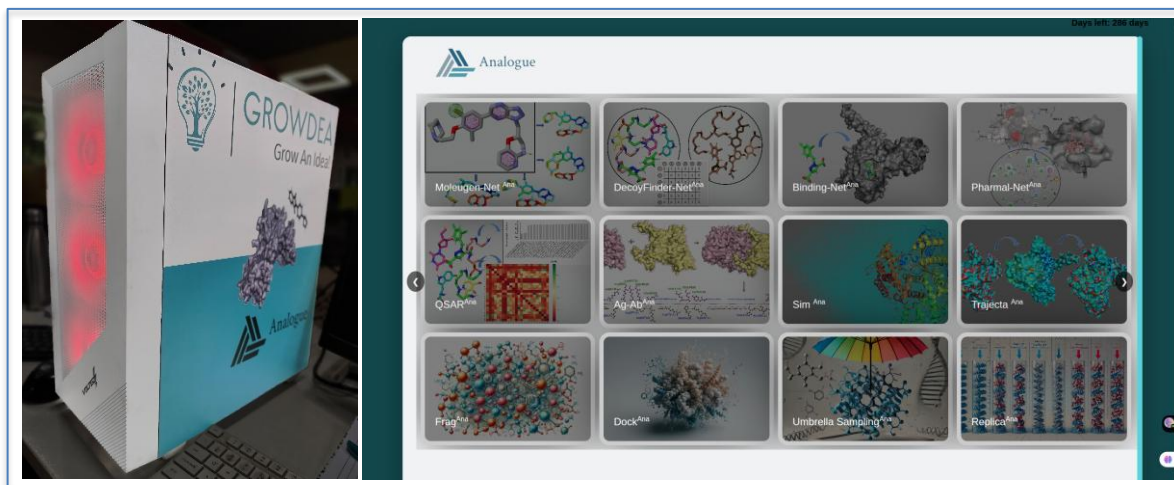




Growdea Therapeutic Platforms

ANALOGUE - 12 Modules

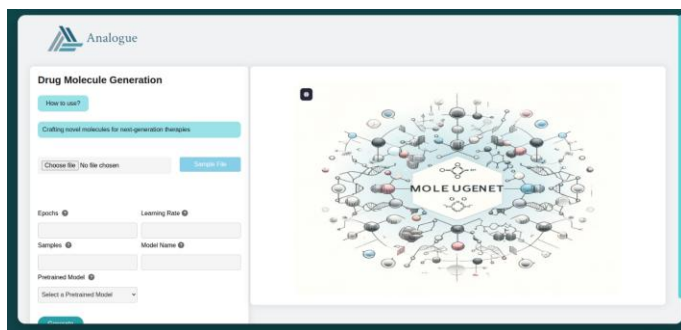


DESCRIPTION

A modular suite for ligand-based prediction, structure-based screening, de novo design, docking, MD simulation, trajectory analysis and free-energy calculations.

KEY CAPABILITIES

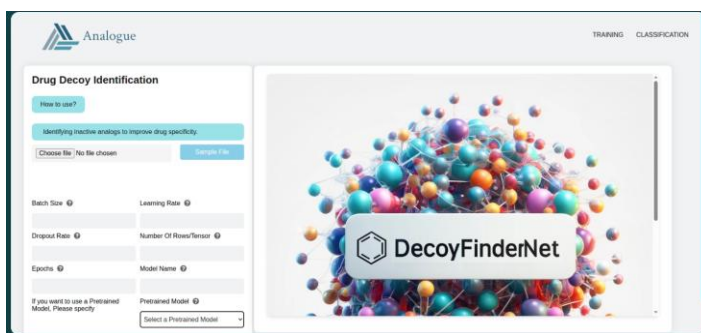
QSAR, De Novo Generation, AI based protein Ligand Binding, Docking, MD Simulation, Replica Exchange, Umbrella Sampling, Trajectory Analysis, Antigen–Antibody interaction analysis.



Module 1: MoleuGen-Net^{Ana}

An AI-driven generative module designed to create novel small molecules. It enables rapid generation of chemically diverse and target-focused compounds with minimal user input. All models were trained on canonical SMILES representations of 271,914 bioactive compounds. This training set was compiled from the

ChEMBL2244 database by retaining compounds with annotated $K_d/IC_{50}/EC_{50} < 1 \mu M$, removing salts and stereo chemical information. This is not specific to protein target and thus this module can generate logical compounds but not specific to any protein target.



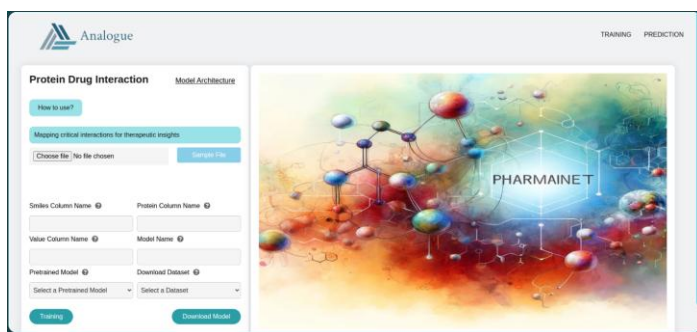
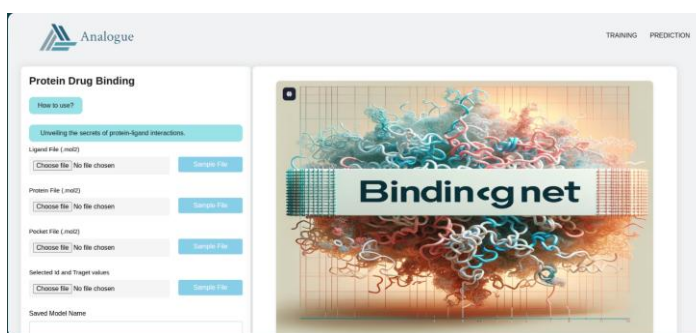
Module 2: Decoy Finder-Net^{Ana}

A machine learning-based classification tool that categorizes small molecules as active or inactive, supporting dataset curation, validation, and model benchmarking for drug discovery workflows. This is using GNN model to classify the compounds as binder and non-binder. This module is specific to

protein, user can give compounds and the model can predict it as binder or non-binder.

Module 3: Binding-Net^{Ana}

Predicts the binding affinity of small molecules against target proteins using advanced AI/ML models, enabling prioritization of high-affinity candidates for further analysis. This is structural method where the module takes docked complex of protein and ligand and predicts the experimental binding affinity as a confirmation of docking.

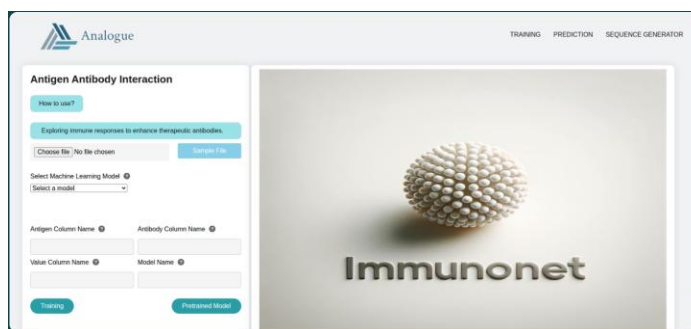


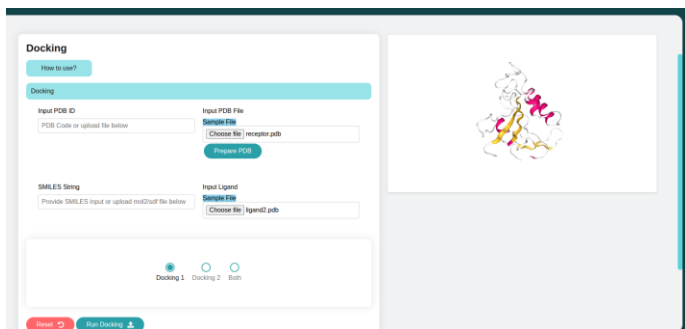
Module 4: Pharma-AI-Net^{ANA}

A sequence-based predictive module that estimates binding affinity directly from protein sequences, eliminating the dependency on resolved 3D structures. This takes the AA sequence of protein and SMILES of compound.

Module 5: Ag-Ab^{Ana}

Specialized module for antibody-antigen interaction optimization, allowing sequence refinement of antibodies or antigens to improve binding specificity and stability.



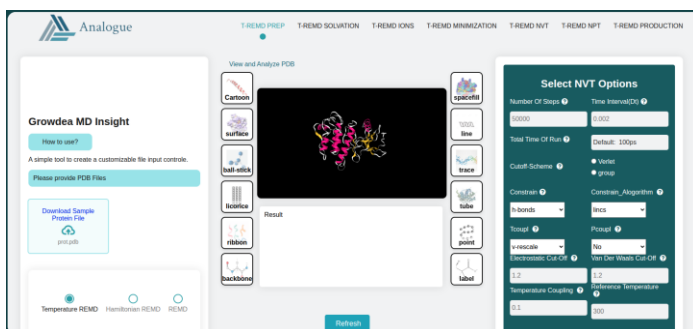
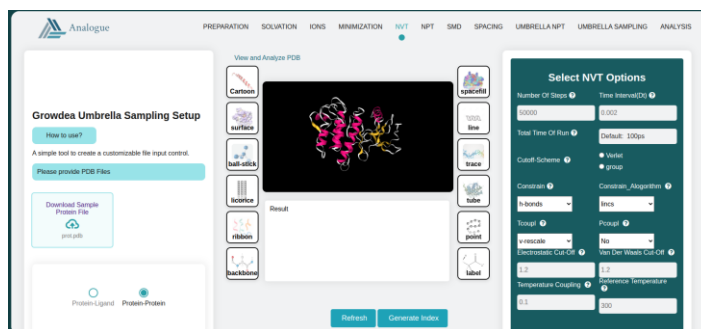


Module 6: Dock^{Ana}

Performs protein-ligand docking using machine learning-assisted and physics-based both approaches via an intuitive graphical interface, enabling accurate pose prediction and interaction analysis.

Module 7: Sim^{ANA} (Molecular Dynamics Simulation)

Provides a complete pipeline for molecular dynamics (MD) simulations with a user-friendly interface, eliminating the need for command-line operations.

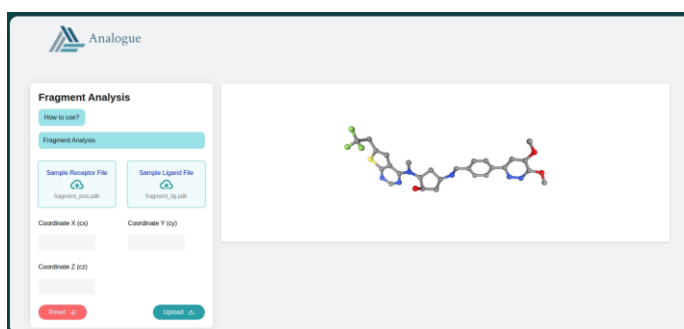


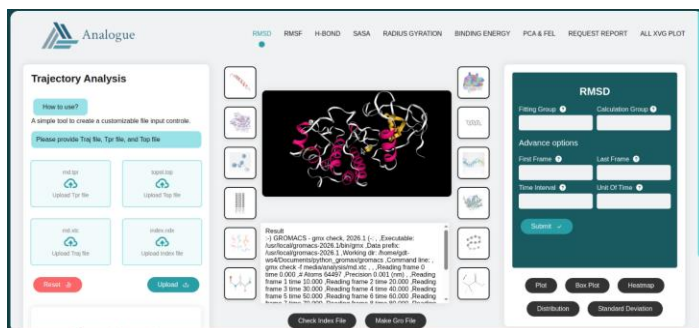
Module 8: Trajecta^{Ana}

An advanced trajectory analysis tool for MD simulations, enabling basic to advanced analyses such as RMSD, RMSF, hydrogen bonding, and conformational dynamics through an interactive UI.

Module 9: Frag^{Ana}

A fragment-based drug design module that generates novel analogue using the main scaffold of the compound using 700000+ fragments, along with predicted binding scores for efficient lead generation.



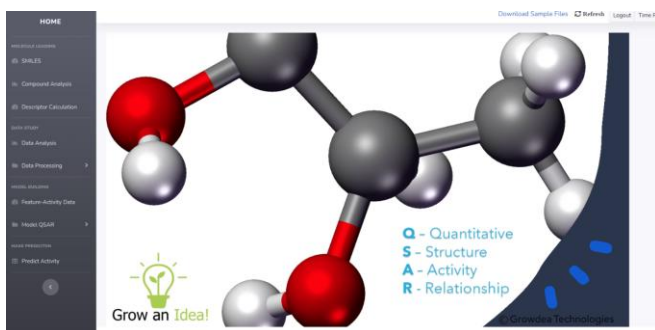
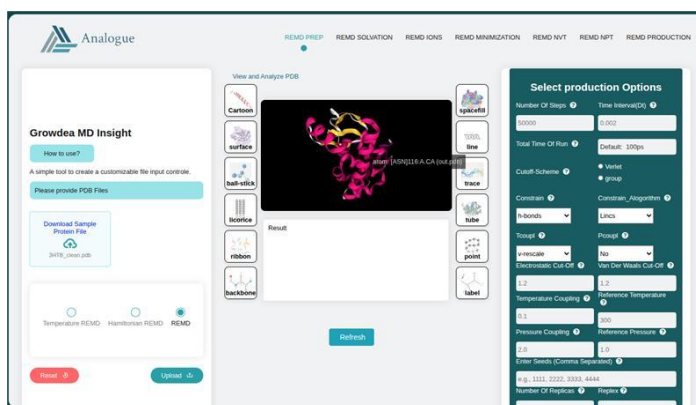


Module 10: Umbrella Sampling

Implements enhanced sampling techniques to overcome energy barriers and explore rarely visited conformational states, enabling accurate calculation of potential of mean force (PMF).

Module 11: Replica Exchange Molecular Dynamics (REMD)

An advanced enhanced sampling technique that performs multiple parallel MD simulations at different temperatures or conditions, allowing periodic exchange between replicas. It improves conformational sampling, helps overcome local minima, and enhances thermodynamic and kinetic predictions.



Module 12: QSAR^{Ana}

A no-code machine learning environment for building QSAR models including descriptor calculation, feature selection, model building, and activity/property prediction. Diverse methods for machine learning for building QSAR model. SMILES of all compounds are

- uploaded. 1500+ chemical descriptors were calculated. ata (Descriptors) was processed {NA removal, Constant/Quasi constant removal, highly correlated removal, normalization}. QSAR model built.



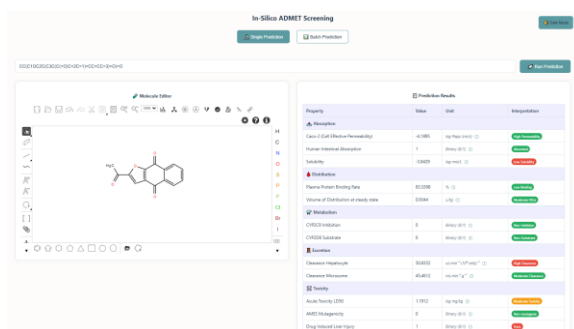
ADMET-Vault - PK Studies

DESCRIPTION

AI-powered in-silico screening for Absorption, Distribution, Metabolism, Excretion, and Toxicity (ADMET), Lead optimization for ADMET

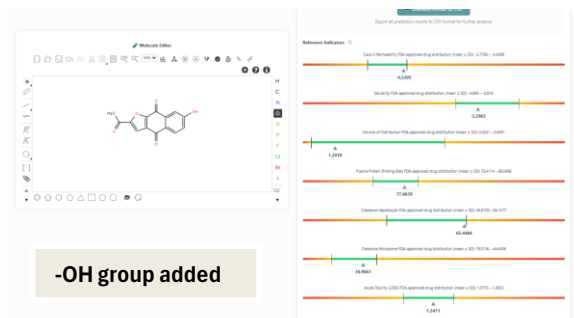
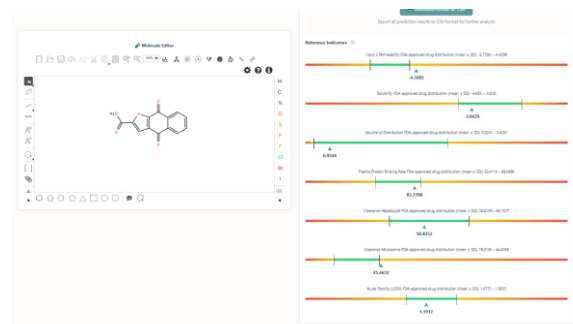
KEY FEATURES

Predicts permeability, protein binding, metabolic liabilities, clearance, and toxicity risk.



A compound was evaluated using the developed ADMET prediction platform to benchmark its pharmacokinetic and toxicity characteristics against FDA-approved drug distributions. The predicted parameters demonstrate strong concordance with known oral drug-like behavior, supporting the reliability of the platform.

This shows only in Caco-2 this drug is slightly outside the range of FDA range which is calculated as mean $\pm$ standard deviation. Similarly, the clearance microsome is also slightly outside the range of approved drugs. ADMET Vault offers to change the molecule and recalculate the ADMET properties to make them under FDA properties range.

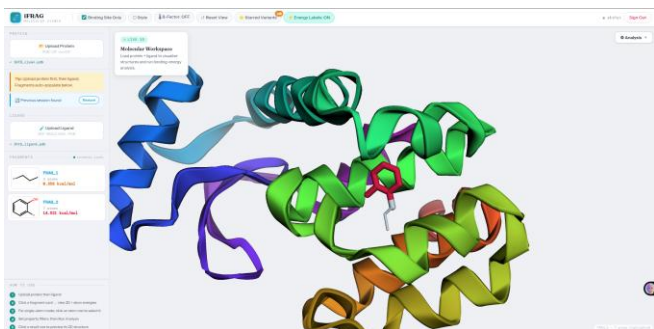


Here, the hydroxyl group (-OH) is added to the compound and the ADMET properties has been changed as shown above. The Caco-2 permeability has decreased to -4.53 but now it is in the range of FDA approved compounds. Moreover, it is observed that solubility has been increased to -3.29 from -3.84 which is due to addition of polar group and remains within

acceptable oral drug-like space. The (Vdss) increased marginally (~ 1.24), suggesting slightly improved tissue penetration. Plasma protein binding ($\approx 77.7\%$) showed a modest reduction compared to the parent profile, consistent with decreased hydrophobic interactions with serum albumin. The predicted ($LD_{50} \approx 1.25$) remained within the acceptable reference range, indicating that -OH functionalization did not introduce new acute toxicity risks.



iFrag - Analogue Generation



DESCRIPTION

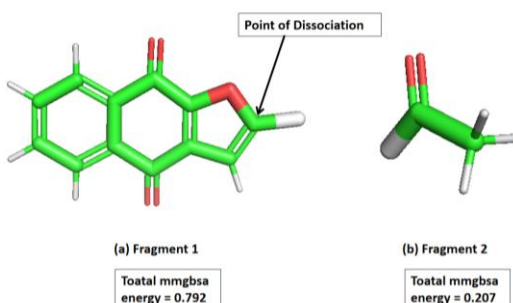
Interaction-guided fragment platform that analyzes docked protein-ligand complexes to identify and replace unfavorable fragments.

KEY FEATURES

Energy minimization, MM/GBSA

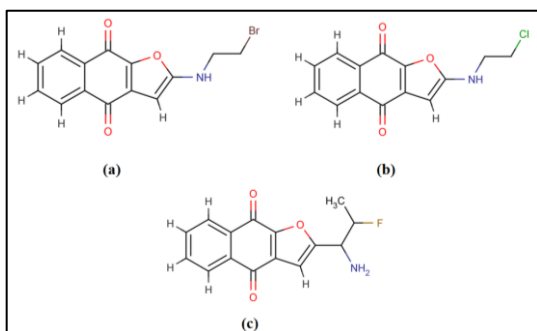
decomposition, fragment replacement proposals.

The tool automatically performed energy minimization to remove steric clashes and optimize the geometry. Normalized energy contribution by heavy atom is shown:



Following per-atom energy decomposition, the ligand was automatically fragmented using **molecular fragmentation algorithms**. This step systematically divided the ligand into chemically meaningful substructures while preserving original atom indices.

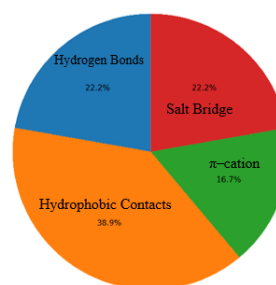
The weak fragment (FRAG_2) was analyzed for nearby protein residues within 6.5 Å. The residues were classified on the basis



of its chemical properties and type of

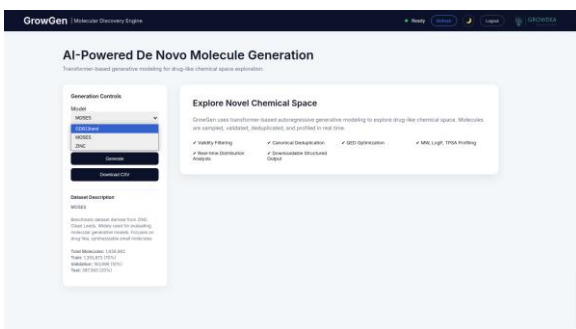
interactions it can make. A fragment library containing 7,08,687 fragments were screened based on the required interaction patterns derived

from the residue environment analysis of the weak fragment.





GrowGen



AI-Powered De Novo Molecule Generation

Transformer-based generative modeling for drug-like chemical space exploration, Generate Novel compounds, drug-like properties, high uniqueness factor, selected compounds screened against target

KEY FEATURES

GrowGen uses transformer-based autoregressive generative modeling to explore drug-like chemical

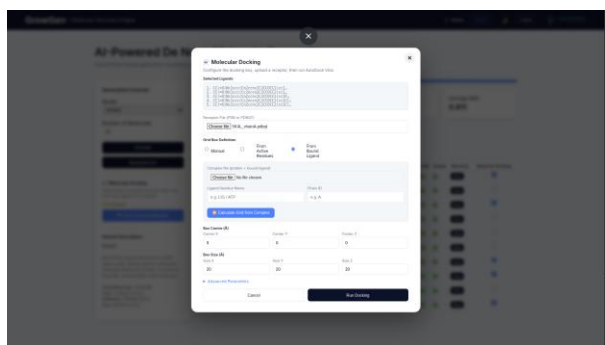
space. Molecules are sampled, validated, deduplicated, and profiled in real time.

- ✓ Validity Filtering
- ✓ Canonical Deduplication
- ✓ QED Optimization
- ✓ MW, LogP, TPSA Profiling
- ✓ Real-time Distribution Analysis
- ✓ Downloadable Structured Output

Generative method deployed, uses Rotary Position Embedding (RoPE) with property conditioning utilizing descriptor embedding and policy-gradient reinforcement learning. Reward scoring was computed using molecular validity, global uniqueness and QED value.

Across several benchmark datasets, including GDB13rand, MOSES, and ZINC, reinforcement learning significantly improves molecular drug-likeness by increasing the average QED while maintaining high levels of chemical validity (above 80%), uniqueness (above 90%), and novelty (above 99%).

ID	SMILES	QED	MW	LogP	TPSA	Valid	Unique	Structure	Select for Docking
1	CC1=NC=CC(=C1)C2=CC=CC=C2C3=CC=CC=C3	0.944	288.27	2.76	90.75	✓	✓	✓	✓
2	CC1=NC=CC(=C1)C2=CC=CC=C2C3=CC=CC=C3	0.944	288.27	2.76	90.75	✓	✓	✓	✓
3	CC1=NC=CC(=C1)C2=CC=CC=C2C3=CC=CC=C3	0.944	288.27	2.76	90.75	✓	✓	✓	✓
4	CC1=NC=CC(=C1)C2=CC=CC=C2C3=CC=CC=C3	0.944	288.27	2.76	90.75	✓	✓	✓	✓
5	CC1=NC=CC(=C1)C2=CC=CC=C2C3=CC=CC=C3	0.944	288.27	2.76	90.75	✓	✓	✓	✓
6	CC1=NC=CC(=C1)C2=CC=CC=C2C3=CC=CC=C3	0.944	288.27	2.76	90.75	✓	✓	✓	✓
7	CC1=NC=CC(=C1)C2=CC=CC=C2C3=CC=CC=C3	0.944	288.27	2.76	90.75	✓	✓	✓	✓
8	CC1=NC=CC(=C1)C2=CC=CC=C2C3=CC=CC=C3	0.944	288.27	2.76	90.75	✓	✓	✓	✓
9	CC1=NC=CC(=C1)C2=CC=CC=C2C3=CC=CC=C3	0.944	288.27	2.76	90.75	✓	✓	✓	✓
10	CC1=NC=CC(=C1)C2=CC=CC=C2C3=CC=CC=C3	0.944	288.27	2.76	90.75	✓	✓	✓	✓



Generate selected number of molecules with high QED score. The tool also allows use to dock the novel generated molecule against the desired protein drug target. Any molecule that are generated could be selected by user and tested against the protein.

Reach Us for More Query:



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Growdea Technologies Pvt Ltd

2026

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